Department of Physics

Please print inside the circle the initial of your **LAST** NAME

Please Print Clearly:

Student:_____ _____

Last Name *First Name*

Student Number: _____ **Student Section:** _____

PCS 125 – Waves and Fields

Mid -Term Test – March 2, 2012

Check the name of your professor:

Dr. Carvalho		Dr. Chithrani		Dr. Tavakkoli		Dr. Yuan	

Read the following attentively

- **The duration of the test is 1 hour and 30 minutes (90 minutes).**
- **The test consists of 3 full-answer questions (4 marks each) and 10 multiple-choice questions (1 mark each). Answer ALL questions.**
- **For each multiple choice question:**
  - **You need to show, in the space provided in this booklet, a justification for your answer.**
  - **The work shown in your exam booklet for the multiple choice questions may be checked at random. If no work or correct argument is shown, the resultant mark will be zero, even if the answer in the bubble sheet is correct.**
  - **Transfer your choices to the bubble sheet only when you're sure of them.**
  - **Some answers may be rounded off.**
- **For the full answer questions, show, in the space provided, all your work. For part marks to be awarded, the steps leading to the answers in each part of the problem have to be organized, unambiguous and clear. DO NOT FORGET UNITS.**
- **All Ryerson examination rules apply.**
- **Sharing of pens, erasers and calculators is not permitted.**
- **Talking to another student or glancing over another student's paper is not permitted and may result in a charge of academic misconduct.**
- **A formula sheet is provided with the test booklet. No other sheets are allowed.**
- **The only calculators allowed are either the Sharp EL-546 or Casio FX-991. If you are found to have another type of calculator (even if not programmable) it will be confiscated for the duration of the test.**

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Student signs here to indicate having read the instructions
(no signature, no mark!)

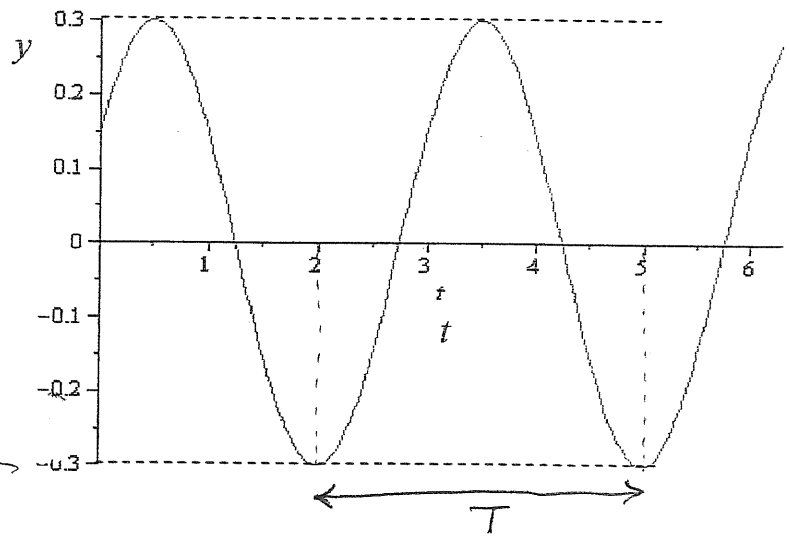
FULL ANSWER QUESTIONS

Question 1. (4 marks)

The plot shown in the figure is a representation of SHM where the vertical axis gives the displacement in centimetres and t is in seconds. At t=0, y= + 0.15 cm.

(a) (2 marks) Extract from the figure the values of the amplitude, frequency, angular frequency and period.

DO NOT FORGET UNITS.



Amplitude: $A = 0.3$ cm

Period: $T = 3$ s

Frequency: $f = \frac{1}{T} = \frac{1}{3} = 0.33$ Hz

Angular frequency: $\omega = 2\pi f$

$= 2\pi \times 0.33 = 2.09$ rad/s

(b) (1 mark) Given the initial condition, determine the value and sign of the phase constant (in rad) if the sinusoidal function is described by a sine function.

$y(t) = A\sin(\omega t + \phi)$

$= 0.3\sin(2.09t + \phi)$

$y(t=0) = 0.15\text{ cm} \Rightarrow 0.15 = 0.3\sin(0+\phi) \Rightarrow \phi = \frac{+\pi}{6} = +0.52$ rad

(c) (1 mark) If the SHM shown above is the source of a transverse traveling wave on a string with mass per unit length 4.0 g/cm and under a tension of 64 N, determine the wavelength of the wave.

$\mu = 4\ g/cm \times \frac{10^{-3}}{10^{-2}} = 0.4\ Kg/m$

$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{64}{0.4}} = 12.65\ m/s$

$\lambda = vT = 12.65 \times 3 = 37.95$ m

Question 2. (4 marks)

A 1.2 W speaker produces a 440 Hz sound radiating equally in all directions in air at 15° C. For the density of air use the value 1.25 kg/m^3.

Determine

(a) (1 mark) the average sound intensity at a distance of 5.0 m

$$I = \frac{P}{4\pi r^2} = \frac{1.2}{4\pi \times (5)^2} = 3.82 \times 10^{-3} \ W/m^2$$

(b) (1 mark) the sound level (loudness) at a distance of 5.0 m

$$\beta = 10 \log\left(\frac{I}{I_0}\right)$$

$$= 10 \log\left(\frac{3.82 \times 10^{-3}}{1 \times 10^{-12}}\right) = 10 \log(3.82 \times 10^{9}) = 95.82 \, dB$$

c) (1 mark) the pressure amplitude at a distance of 5.0 m

$$I = \frac{\Delta P_{max}^2}{2\rho v} \Rightarrow \Delta P_{max} = \sqrt{2 I \rho v}$$

$$v = 331\sqrt{1 + \frac{T_c}{273}} = 331\sqrt{1 + \frac{15}{273}} \simeq 340 \ m/s$$

$$\Delta P_{max} = \sqrt{2 \times 1.25 \times 340 \times 3.82 \times 10^{-3}} = 1.80 \ Pa$$

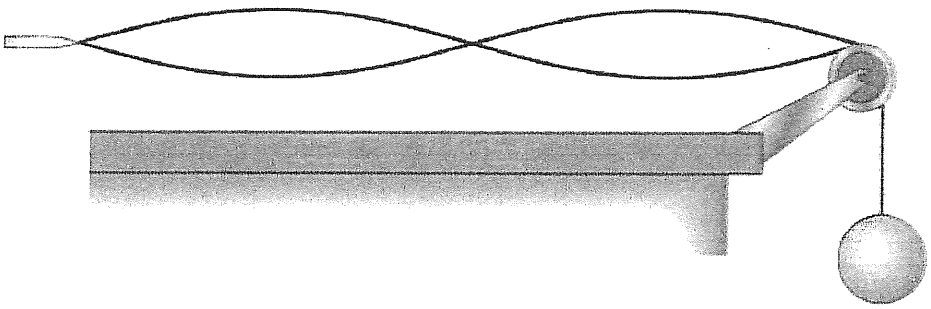
d) (1 mark) the displacement amplitude at a distance of 5.0 m.

$$\Delta P_{max} = \rho v \omega S_{max} \Rightarrow S_{max} = \frac{\Delta P_{max}}{\rho v \omega}$$

$$S_{max} = \frac{1.80}{1.25 \times 340 \times 2\pi \times 440} = 1.53 \times 10^{-6} \ m = 1.53 \ \mu m$$

Question 3. (4 marks)

One end of a horizontal string is attached to a vibrating blade with frequency 84 Hz and the other end passes over a pulley. A sphere of mass 2.00 kg hangs on the other end of the string. The mass per unit length of the string is 56 g/m. The string is vibrating in its second harmonic.



(a) (0.5 mark) Determine the length of the string on which the standing wave is formed.

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{2 \times 9.81}{56 \times 10^{-3}}} = 18.72 \text{ m/s}$$

$$\lambda = \frac{v}{f} = \frac{18.72}{84} = 0.22 \text{ m}$$

Second harmonic $\Rightarrow$ $L = \lambda = 0.22$ m

(b) (2 marks) Assume that both the frequency of the vibrating blade and the length of the string remain the same. Determine the largest mass the sphere can have so that a standing wave can still be formed on the string.

Largest mass corresponds to the fundamental viberation :

$$\lambda = 2L = 2 \times 0.22 = 0.44 \text{ m}$$

$$v = \lambda f = 0.44 \times 84 = 36.96 \text{ m/s}$$

$$T = \mu v^2 = 56 \times 10^{-3} \times (36.96)^2 = 76.50 \text{ N} \Rightarrow m = \frac{T}{g} = \frac{76.50}{9.81} = 7.80 \text{ kg}$$

(c) (1.5 mark) Knowing that the amplitude of the loops is 5 cm, write the mathematical equation for the standing wave with the two loops as shown in the figure above.

$$\omega = 2\pi f = 2\pi \times 84 = 168\pi \text{ rad/s}$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{0.22} = 9.09\pi \text{ rad/s}$$

$$2A = 2 \times 0.05 = 0.1 \text{ m}$$

$$y(x,t) = 2A(\sin kx)(\cos \omega t) = 0.1 \sin(9.09\pi x)(\cos 168\pi t)$$

MULTIPLE CHOICE QUESTIONS

Question 1.

Object A is four times heavier than object B. Each object is attached to a spring, and both springs have the same spring constant. Both objects are then pulled from their equilibrium position and released from rest.

What is the ratio of the periods of the two oscillators if the amplitude of A is half the amplitude of B.

a) $T_A/T_B=2$ b) $T_A/T_B=1/4$ c) $T_A/T_B=4$ d) $T_A/T_B=1/2$

e) other (give the value)

Question 2.

A certain spring elongates 9.0 mm when it is suspended vertically and a block of mass *M* is hung on it. If driven by a force of variable frequency, at what frequency would resonance be obtained?

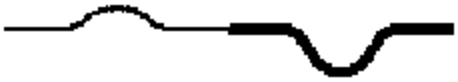
a) 0.088 rad/s b) 200 rad/s c) 33 rad/s d) 1140 rad/s
e) cannot be computed unless the value of *M* is given

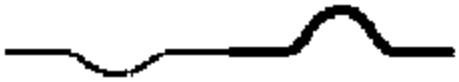
Question 3.

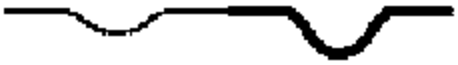
Two ropes are spliced together as shown and a pulse is traveling to the right on them.

A short time after the incident pulse shown in the diagram above, reaches the splice, what will be the rope's appearance?

a)

b) 

c) 

d) 

e)

Question 4.

A transverse wave traveling on a cord in the negative *x* direction has a wave number of 60 rad/m, a period of 0.20 s and an amplitude of 3.0 mm. What is the maximum transverse acceleration of a point on the cord

a) 180 mm/s^2 b) 60 mm/s^2 c) 4.87 m/s^2

d) 2.96 m/s^2 e) other (give the value)

Question 5.

A sinusoidal wave is traveling on a string with velocity 40 cm/s. The displacement of the particles of the string at x=10 cm is given by the equation y=(5.0) sin[1.0π–(4.0π) t]. where y is given in centimeters and t in seconds. Determine the wavelength of the wave.

a) 20.0 cm b) 10.0 cm c) 3.14 cm

d) 0.314 m e) other (give the value)

Question 6.

A truck moving at 36 m/s passes a police car moving at 45 m/s in the opposite direction. If the frequency of the siren relative to the police car is 500 Hz, what is the frequency heard by an observer in the truck after the police car passes the truck? (The speed of sound in air is 343 m/s.)

a) 636 Hz

b) 396 Hz

c) 361 Hz

d) 393 Hz

e) 617 Hz

Question 7.

A wave generated in a medium is a longitudinal wave when

a) there is a net transport of matter by the wave.

b) the molecules of the medium are unable to exert forces on each other.

c) molecular displacements are parallel to the wave velocity.

d) molecular displacements are perpendicular to the wave velocity.

e) the density of the medium is less than the density of water.

Question 8.

A trumpet player hears 3 beats per second when she plays a note and simultaneously sounds a 440 Hz tuning fork. After pulling her tuning valve out to slightly increase the length of her trumpet, she hears 5 beats per second against the tuning fork. What was her initial frequency?

a) 437 Hz

b) 443 Hz

c) Either a) or b).

d) 435 Hz

e) 445 Hz

Question 9.

A vertical tube one meter long is open at the top. It is filled with 75 cm of water. If the velocity of sound is 344 m/s, what will the fundamental resonant frequency be (in Hz)?

a) 3.4

b) 172

c) 344

d) 1.7

e) 688

Question 10.

Two sinusoidal waves are described by $y_1 = 3\sin(94x - 700t)$ and $y_2 = 3\sin(94x + 700t)$ where the displacements y_1 and y_2 are measured in metres. What is the amplitude of the resultant wave when they interfere?

a) 8.0 m

b) 4.3 m

c) 3.0 m

d) 2.0 m

e) 6.0 m